SmartBiz AI

n8n Automation User Manual

Module: Workflow Automation
Responsible Member: Member 2 — n8n Automation Engineer
Platform: n8n
Backend: FastAPI
Communication: REST API, HTTP Request, JSON
Project: SmartBiz AI

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1. Introduction

SmartBiz AI is an AI-powered business automation system designed to reduce manual work and improve business productivity.

The n8n automation module connects the frontend, backend APIs, email services, and notification system through automated workflows.

This module contains three main workflows:

1. Lead Automation
2. Customer Follow-up
3. Notification Workflow

2. Technology Used

Technology	Purpose
n8n	Workflow automation
FastAPI	Backend API
HTTP Request	API communication
Webhook	Receive external requests
Gmail	Send emails
JSON	Data exchange
Ngrok	Temporary public access to local backend

3. Lead Automation Workflow

Purpose

To automatically receive new lead information, process the data, save it to the backend, create a notification, and send a welcome email.

Workflow Structure

Webhook → Edit Fields → IF → Backend and Notification/Email Operations

Node Description

1. Webhook

Receives lead information from an external form or application through an HTTP request.

2. Edit Fields

Processes and organizes the incoming lead data into the required structure.

3. IF

Checks whether the required conditions are satisfied before continuing.

4. Save Lead to Backend

Sends lead information to the FastAPI backend through an HTTP request.

5. Create Notification

Creates a notification for the newly received lead.

6. Generate Welcome Email

Prepares the welcome email content.

7. Send a Message (Gmail)

Sends the welcome email through the configured Gmail account.

8. Update Email Log Status

Updates the email delivery status in the backend.

Expected Outcome

- Lead information is received.
- Lead data is saved in the backend.
- A notification is created.
- A welcome email is sent.
- Email status is recorded.

4. Customer Follow-up Workflow

Purpose

To identify leads requiring follow-up, send follow-up emails, and record follow-up activities in the backend.

Workflow Structure

Schedule Trigger → HTTP Request → Split Out → IF → Gmail → HTTP Request

Node Description

1. Schedule Trigger

Starts the workflow according to a configured schedule.

2. HTTP Request

Retrieves due follow-up leads from the backend.

API endpoint:

```
GET /api/follow-ups/due?days=3
```

3. Split Out

Separates the returned lead collection into individual items for processing.

4. IF

Checks whether the lead has the required email address and lead ID.

5. Gmail

Sends a follow-up email to the selected lead.

6. HTTP Request

Records the follow-up activity in the backend.

API endpoint:

```
POST /api/follow-ups
```

Example request body:

```
{
  "lead_id": 7,
  "channel": "email",
  "status": "sent",
  "message": "Follow-up email sent via Gmail"
}
```

Duplicate Prevention

The backend prevents duplicate follow-up records within the configured restriction period.

During integration testing, a repeated follow-up request returned:

```
409 Conflict
```

This confirms that the duplicate prevention mechanism was working during the test.

Expected Outcome

- Due leads are retrieved.
- Each lead is processed individually.
- Eligible leads receive follow-up emails.
- Follow-up activities are recorded in the backend.
- Duplicate requests are rejected by the backend.

5. Notification Workflow

Purpose

To create notifications automatically when a notification event is received.

Workflow Structure

Webhook → Edit Fields → HTTP Request

Node Description

1. Webhook

Receives notification-related information from an external request.

2. Edit Fields

Creates the notification message using the incoming data.

Example expression:

```
{{ "New lead from website: " + $json.body.name }}
```

3. HTTP Request

Sends the generated notification message to the FastAPI backend.

API endpoint:

```
POST /api/notifications
```

Example request body:

```
{
  "message": "New lead from website: SmartBiz Test Lead"
}
```

Notification API Testing

The following backend API operations were tested:

Method	Endpoint	Purpose
POST	`/api/notifications`	Create notification
GET	`/api/notifications`	Retrieve notifications

PATCH	`/api/notifications/{id}/read`	Mark notification as read

The POST, GET, and PATCH operations returned successful responses during testing.

Why Is There No Switch Node?

The latest Notification workflow uses only three nodes because the required notification creation process does not need multiple conditional branches.

6. Backend Integration

The n8n workflows communicate with the FastAPI backend through HTTP requests.

The integration uses:

- HTTP methods such as GET, POST, and PATCH.
- JSON request and response data.
- Appropriate request headers.
- Backend API endpoints.
- Ngrok for temporary public access to a locally hosted backend during integration.

Required headers used for the Ngrok connection:

```
Content-Type: application/json
ngrok-skip-browser-warning: true
```

The Content-Type header is used for JSON request bodies. The Ngrok header is used when required by the tunnel configuration.

7. Testing and Verification

The following integration tests were completed:

Workflow	Testing Activities
Lead Automation	Lead saving, notification creation, welcome email and email logging
Customer Follow-up	Due lead retrieval, item splitting, email sending and duplicate prevention
Notification	POST, GET and PATCH API operations

Important Test Results

- Lead Automation successfully saved lead and related records during testing.
- Customer Follow-up successfully created a follow-up record.
- Duplicate follow-up request returned HTTP 409 Conflict.
- Notification creation returned a new notification ID.
- Notification retrieval returned notification records.
- PATCH successfully changed a notification's read status to true.

8. Common Troubleshooting

Problem	Possible Cause	Action
Webhook does not receive data	Incorrect URL or inactive workflow	Verify webhook configuration and workflow status
HTTP Request fails	Backend unavailable	Check backend and tunnel availability
404 Not Found	Incorrect API endpoint	Verify endpoint path
422 Unprocessable Entity	Invalid request data	Check JSON fields and data types
409 Conflict	Duplicate follow-up request	Check existing follow-up record
Gmail fails	Gmail credential or configuration issue	Verify connected Gmail credentials
Ngrok endpoint offline	Tunnel stopped	Coordinate with backend administrator

9. Workflow Maintenance

1. Keep the latest workflow JSON files in the project GitHub repository.
2. Export workflows after approved changes.
3. Keep API URLs updated when the backend URL changes.
4. Verify credentials before running email workflows.
5. Review execution history when investigating errors.
6. Coordinate workflow activation with the backend administrator during database resets and demonstrations.

10. Conclusion

The n8n automation module connects the SmartBiz AI backend with automated lead management, customer follow-up, and notification processes.

These workflows reduce repetitive manual tasks and support more efficient business operations.

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Project: SmartBiz AI